**Skill Development**

**Project Report On**

# HAWK-EYE(Smart gateway with Wi-Fi monitoring)

A Dissertation submitted to JNTU Hyderabad in partial fulfillment of the academic requirements for the award of the degree.

## Bachelor of Technology

## in

## Computer Science And Engineering(Cyber Security)


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Giving Wings To Thoughts

St. PETER'S

ENGINEERING COLLEGE

(UGC - AUTONOMOUS)

**Department Of Computer Science And Engineering**

**(Cyber Security)**

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**2024- 2028**

# ST. PETER'S ENGINEERING COLLEGE

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## CERTIFICATE

This is to certify that the Skill Development Project I report entitled **" HAWK-EYE(Smart gateway with Wi-Fi monitoring)"** being submitted by K. Likhith Bhanu (24BK1A6228), CH. Trinath (24BK1A6209) in partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering(Cyber Security)** is a record of Bonafide work carried out his/her under my guidance and supervision.

The results embodies in this project report have not been submitted to any other University or Institute for the award of any Degree.

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# ABSTRACT

This project presents the design and implementation of a smart Wi-Fi gateway system named *Hawk-Eye*, developed using the ESP32 microcontroller. The system aims to monitor and manage devices connected to a local wireless network in real time. Unlike traditional Wi-Fi scanners that passively observe nearby networks, this system creates a controlled access point through which devices must connect, enabling precise and reliable monitoring. The ESP32 operates in dual mode, functioning simultaneously as a station and an access point. It connects to an upstream network such as a mobile hotspot while also broadcasting its own network for client devices. This architecture allows the system to act as an intermediary node, providing a structured environment for device tracking. The core functionality includes identifying devices based on their MAC addresses and maintaining a dynamic record of connected clients. The system is capable of detecting newly connected devices, recognizing previously known devices upon reconnection, and identifying when devices disconnect from the network. This continuous monitoring is performed in real time, ensuring accurate status updates. To improve reliability in dynamic environments, the system incorporates a retry-based connection mechanism that handles unstable network conditions effectively. This makes it suitable for use in crowded areas where multiple wireless networks may interfere with connectivity. The project also considers hardware limitations of the ESP32, particularly in implementing full network address translation. While complete routing functionality is constrained, the system successfully demonstrates gateway-level monitoring capabilities without compromising stability. Hawk-Eye provides a low-cost and efficient solution for basic network surveillance and device tracking applications. It can be extended for use in smart environments, security monitoring, and educational demonstrations of embedded networking concepts. The project highlights the practical integration of networking and embedded systems to build an intelligent monitoring platform.

# CHAPTER 1
# INTRODUCTION

# CHAPTER 1

# INTRODUCTION

## 1.1. Problem Statement

With the rapid growth of wireless communication, modern environments are saturated with multiple Wi-Fi networks and connected devices. However, monitoring which devices are actively connected to a specific network in real time remains a challenge. Conventional tools either scan all nearby networks without precision or require complex configurations that are not suitable for lightweight or embedded systems. This often leads to difficulty in identifying unauthorized connections, tracking device activity, or understanding network usage patterns. In crowded environments such as colleges or public spaces, interference and multiple access points further complicate monitoring. There is a clear need for a compact and efficient solution that can create a controlled network environment and accurately track connected devices. This project addresses that need by developing a smart Wi-Fi gateway system that enables real-time monitoring of devices within a dedicated network using an embedded platform. .

## 1.2. Research Objective

The primary objective of this project is to design and implement a lightweight smart Wi-Fi gateway using the ESP32 microcontroller that can monitor connected devices in real time. The system aims to create a controlled wireless network through which client devices connect, allowing precise identification and tracking using MAC addresses. It focuses on detecting newly connected devices, recognizing previously known devices upon reconnection, and identifying when devices leave the network. Another objective is to enable dual-mode operation, where the system connects to an external network while simultaneously hosting its own access point, thereby acting as an intermediary monitoring node. The project also emphasizes stability in dynamic environments by incorporating retry mechanisms to handle unstable connections. Additionally, it aims to provide a simple and modular architecture that can be extended for advanced applications such as network security monitoring, smart environments, and educational demonstrations. The final goal is to present a reliable and cost-effective solution that bridges embedded systems and basic network management concepts.

## 1.3 Project Scope and Limitations

### Scope

This project focuses on developing a compact and efficient Wi-Fi monitoring system using the ESP32 microcontroller with dual-mode capability. The system is designed to operate as both an access point and a station, enabling it to create a controlled network while maintaining connectivity to an external source. The scope includes: Creating a dedicated wireless access point to which client devices such as smartphones, laptops, or other Wi-Fi-enabled devices can connect for controlled monitoring. Connecting the ESP32 to an external network such as a mobile hotspot or router to simulate gateway functionality and demonstrate real-world networking behavior. Monitoring connected devices in real time by capturing and processing their MAC addresses, ensuring accurate identification of each device.

## Limitations

The system monitors only devices connected to its own access point and does not scan all nearby networks, which limits its visibility to a controlled environment only. Full internet routing using network address translation is limited due to hardware and framework constraints of the ESP32, making it unsuitable as a complete replacement for a traditional router. The system identifies devices using MAC addresses only and does not perform deep packet inspection or content analysis, which restricts its ability to analyze actual data traffic. Performance may vary in highly congested wireless environments due to interference, signal overlap, and channel limitations, especially in crowded areas with multiple active networks. The number of devices that can be tracked is limited by the memory and processing constraints of the microcontroller, which may affect scalability in larger deployments. Advanced threat detection techniques such as machine learning, behavioral analysis, or anomaly detection are not implemented in this version, reducing its capability to identify complex or hidden threats. The system does not maintain long-term persistent logs unless externally integrated, which limits historical data analysis and auditing capabilities. Power supply stability and hardware quality can influence performance, especially during continuous operation over extended periods. The system relies on proper network configuration, and incorrect hotspot or router settings may affect connectivity and overall functionality. The system is intended for educational and authorized use only and should not be deployed for unauthorized monitoring, as misuse may lead to ethical and legal issues.

# CHAPTER 2
# BACKGROUND WORK

# CHAPTER 2

# BACKGROUND WORK

## 2.1 Basic Wi-Fi Monitoring Systems

### 2.1.1 Introduction

Basic Wi-Fi monitoring systems are among the simplest approaches used to observe and analyze wireless network activity. These systems operate by scanning available networks or tracking devices connected to a specific access point. The primary purpose of such systems is to identify connected devices, monitor their presence, and understand network usage patterns. Unlike advanced network analyzers, basic monitoring systems focus on capturing essential information such as MAC addresses, signal strength, and connection status without deep inspection of data traffic.Most basic Wi-Fi monitoring systems rely on the wireless capabilities of hardware modules and built-in networking libraries to detect and process connection events. These systems may function in passive mode, where nearby networks are scanned, or in active mode, where a dedicated access point is created to control and monitor device connections. Over time, these systems have become important tools in network management, educational demonstrations, IoT applications, and entry-level cybersecurity projects. Although simple in design, they provide foundational insights into how wireless communication works and how devices interact within a network. They also serve as a base for more advanced systems that incorporate traffic analysis, intrusion detection, and intelligent monitoring techniques. Their simplicity makes them highly suitable for embedded platforms like microcontrollers, where resource efficiency and real-time performance are critical.

### 2.1.2 Merits, Demerits and Challenges

#### Merits

Simple and Lightweight: Requires minimal hardware resources and can be implemented on low-power devices like microcontrollers.Real-Time Monitoring: Provides instant updates on device connections and disconnections within the network.Easy Deployment: Can be quickly set up without complex configurations or external dependencies. Device Identification: Uses MAC addresses to uniquely identify connected devices. Educational Value: Helps in understanding wireless networking concepts and embedded system integration. Flexible Integration: Can be extended with additional modules such as alerts, logging, or visualization tools.

#### Demerits

Limited Visibility: Monitors only devices connected to a specific network and not all nearby wireless activity. No Deep Analysis: Does not inspect packet data or analyze actual network traffic content. Dependence on Hardware: Performance is restricted by the capabilities of the microcontroller

## Challenges

Network Interference: Performance may be affected in environments with multiple overlapping Wi-Fi signals. Channel Management: Ensuring proper communication when multiple networks operate on different channels. Connection Stability: Maintaining reliable connections in dynamic or crowded environments. Resource Constraints: Managing memory and processing limits of embedded systems like ESP32. Device Re-identification: Accurately tracking devices when they disconnect and reconnect frequently. Compatibility Issues: Handling variations in hotspot settings, security protocols, and network configurations.

## Implementation of Basic Keystroke Logging Systems

Basic Wi-Fi monitoring systems are typically implemented using microcontrollers such as the ESP32 along with built-in networking libraries. The implementation involves configuring the device to operate in dual mode, allowing it to function as both an access point and a station. This enables the system to create its own network while maintaining connectivity to an external source. The general implementation flow includes initializing the Wi-Fi module and setting up a dedicated access point for client devices. The system then continuously scans for connected devices and retrieves their MAC addresses using available network APIs. These addresses are processed and stored in a structured format to maintain a record of active and previously connected devices. The system compares current and previous device lists to detect new connections, reconnections, and disconnections in real time. Additional mechanisms such as retry-based connection handling are used to ensure stability when connecting to external networks like mobile hotspots. The monitored data is displayed through a serial interface in a clear and organized manner for observation and analysis. Care is taken to optimize memory usage and maintain efficient processing, ensuring the system runs smoothly without interruptions. Overall, the implementation focuses on simplicity, reliability, and real-time responsiveness while providing a strong foundation for further enhancements such as alert systems, data logging, or integration with advanced network monitoring frameworks.

# CHAPTER 3
# PROPOSED SYSTEM

# CHAPTER 3

# PROPOSED SYSTEM

## 3.1 Objective of Proposed Model

The primary objective of the proposed Hawk-Eye Wi-Fi Gateway system is to provide a structured and efficient mechanism for monitoring and managing devices connected to a wireless network in real time. The system is designed to create a dedicated access point, capture device connection details, and generate meaningful insights based on network activity. By enabling controlled connectivity, the model allows accurate identification of devices using their MAC addresses and supports continuous tracking of their connection status. The system is capable of detecting new device connections, recognizing previously known devices upon reconnection, and identifying disconnection events, thereby maintaining a dynamic and updated network view. The proposed model emphasizes reliability in monitoring, efficiency in handling multiple devices, and stability in dynamic environments such as crowded wireless spaces. It incorporates mechanisms to maintain consistent connectivity with external networks while simultaneously hosting its own access point, ensuring uninterrupted operation. Additionally, the system focuses on simplicity, low resource utilization, and ease of deployment, making it suitable for embedded platforms like the ESP32. The design is modular, allowing future enhancements such as alert systems, data logging, or integration with advanced network analysis tools. The objective extends beyond basic device detection by enabling structured observation of network behavior, which can be useful for security monitoring, usage analysis, and educational demonstrations. Overall, the proposed model aims to combine real-time data acquisition with efficient device tracking to create a practical and scalable wireless monitoring solution.

## 3.2 Algorithms Used for Proposed Model

The proposed system utilizes a combination of real-time monitoring logic and comparison-based algorithms to track and analyze device connectivity within the network environment. These algorithms ensure accurate detection of device states and efficient handling of network data.

### Device Monitoring Algorithm

The system continuously monitors devices connected to the access point using built-in Wi-Fi functions of the ESP32. It retrieves the list of currently connected devices and extracts their MAC addresses. Each detected device is stored in a temporary list representing the current network state. The system compares this list with a previously stored list of devices to determine changes in connectivity. If a device appears in the current list but not in the previous list, it is identified as a newly connected device. If the device was previously known, it is marked as a reconnection. This comparison process ensures accurate real-time tracking of network activity. The algorithm runs in a loop with a fixed interval, allowing continuous updates while maintaining system stability. Efficient memory handling is applied to store and update device lists without exceeding hardware limitations.

## Device State Detection Algorithm

This algorithm focuses on identifying the status of devices within the network. By comparing current and previous device lists, the system detects three main states: new connections, reconnections, and disconnections. When a device is present in the current list but not in the previous list, it is classified as a new or reconnected device depending on its history. If a device was previously connected but is no longer found in the current list, it is marked as disconnected. This method ensures that all device transitions are captured accurately, providing a clear understanding of network activity over time. The results are displayed in a structured format through the serial interface for easy interpretation.

## Connection Stability Algorithm

To ensure reliable operation, the system implements a retry-based connection mechanism for connecting to external networks such as mobile hotspots or routers. If the connection attempt fails, the system automatically retries after a short delay until a successful connection is established. This approach improves performance in unstable or congested environments where network availability may fluctuate. It ensures that the monitoring system remains active and continues functioning without manual intervention.

## 3.3 Designing

The design of the proposed Hawk-Eye Wi-Fi Gateway system focuses on modularity, real-time monitoring, and efficient device tracking. The system is structured into multiple functional modules to ensure clarity, scalability, and ease of maintenance. It is designed to operate on the ESP32 microcontroller using its dual-mode Wi-Fi capability, enabling both access point and station functionalities simultaneously. The first module is the **Access Point Module**, which creates a dedicated wireless network for client devices. This module allows controlled connectivity, ensuring that only devices connected to this network are monitored. It provides the foundation for capturing device information in a structured manner. The second module is the **Device Monitoring Module**, which continuously scans for connected devices and retrieves their MAC addresses. This module maintains a dynamic list of devices and updates it in real time. It ensures efficient tracking of device activity without causing performance issues. The third module is the **Device State Analysis Module**, which compares current and previous device lists to identify connection events. It detects new devices, recognizes reconnections, and identifies disconnections, providing a clear view of network behavior. The final module is the **Output and Reporting Module**, which displays structured information through the serial monitor. It ensures that all detected events are presented in a readable format for analysis and demonstration purposes. The overall design emphasizes low resource usage, stability in dynamic environments, and the ability to extend the system with additional features such as alerts, logging, or integration with advanced monitoring tools.

### 3.3.1 UML Diagram

The UML diagram for the system can be represented as:

**Classes:-**

**Access Point Manager** → Responsible for creating and managing the Wi-Fi network hosted by the ESP32.

**Device Monitor** → Detects connected devices and retrieves their MAC addresses in real time.

**Device Analyzer** → Compares device lists to identify new connections, reconnections, and disconnections.

**Output Manager** → Formats and displays device information in a structured and readable form.

**Interactions**:-

The Access Point Manager establishes the network and allows devices to connect. The Device Monitor continuously collects data about connected devices and forwards it to the Device Analyzer. The Device Analyzer processes the data, identifies device states, and determines changes in connectivity. The Output Manager receives processed information and presents it to the user through the serial interface. The design ensures clear separation of responsibilities, making the system easy to maintain, debug, and extend for future enhancements such as alert systems or advanced analytics.

## 3.3 Stepwise Implementation and Code

The implementation of the Hawk-Eye Wi-Fi Gateway system begins with configuring the ESP32 microcontroller to operate in access point mode, where a dedicated wireless network is created using a predefined SSID and password, allowing client devices to connect in a controlled environment. Once the access point is successfully initialized, the system is further configured to operate in station mode simultaneously, enabling it to connect to an external network such as a mobile hotspot or router, thereby establishing a gateway-like structure. A retry-based connection mechanism is implemented to ensure reliable connectivity in dynamic or unstable environments, where the ESP32 continuously attempts to reconnect until a valid IP address is obtained. After establishing both access point and station functionalities, the system proceeds to the device monitoring phase, where it retrieves the list of connected devices using built-in Wi-Fi APIs provided by the ESP32 framework. The MAC addresses of all connected devices are extracted and stored in a temporary list representing the current network state. This list is then compared with a previously stored list of devices to identify changes in connectivity, allowing the system to detect new device connections, recognize previously known devices upon reconnection, and identify devices that have disconnected from the network. The system maintains a dynamic record of known devices to ensure accurate tracking over time. All detected events are processed and displayed through the serial monitor in a structured format, clearly indicating the status of each device for easy observation and analysis. The entire process runs in a continuous loop with controlled delays to ensure real-time monitoring while maintaining system stability and efficient resource usage.

# CHAPTER 4
# RESULTS AND DISCUSSION

# CHAPTER 4

# RESULTS AND DISCUSSION

The analysis module of the Hawk-Eye Wi-Fi Gateway system processes real-time device connectivity data to generate meaningful insights into network activity and behavior. The system continuously monitors devices connected to the access point and records their presence, allowing identification of patterns such as frequent connections, disconnections, and device reappearances. This enables the system to highlight both normal usage trends and unusual connectivity behavior. The structured output provides a clear understanding of how devices interact with the network over time, making it easier to detect irregular activity. By maintaining a dynamic record of device states, the system ensures accurate and consistent monitoring, even in environments with frequent connection changes.

## 4.3.0 Connected Device Monitoring

The system continuously tracks all devices connected to the network and displays their MAC addresses in real time. Each device is uniquely identified, allowing clear differentiation between multiple clients connected to the access point. This monitoring provides a complete view of active devices at any given moment, helping in understanding network usage patterns.The structured output enables easy observation of how many devices are connected simultaneously and how frequently devices join or leave the network. This forms the foundation for further analysis and helps establish a baseline of normal network activity.

## 4.3.1 New Device Detection

The system automatically identifies newly connected devices that have not been previously recorded. When a new device connects, it is flagged and displayed with a clear indication, allowing immediate recognition. This feature is particularly useful in detecting unauthorized or unexpected devices attempting to access the network. By maintaining a list of known devices, the system ensures that only first-time connections are marked as new, improving accuracy and reducing false alerts. This enhances the system's ability to provide real-time awareness of network changes.

## 4.3.2 Device Reconnection Detection

Devices that disconnect and later reconnect to the network are identified as reconnected devices. The system compares current device data with previously stored records to determine whether a device has been seen before. This functionality helps in distinguishing between completely new devices and returning devices, providing better insight into user behavior. It also allows tracking of device activity over time, which can be useful in identifying repeated access patterns or intermittent connectivity issues.

## 4.3.3 Device Disconnection Detection

The system detects when a device is no longer connected to the network by comparing current and previous device lists. If a previously connected device is not found in the current scan, it is marked as disconnected.

### 4.3.4 Alert Categorization

The Hawk-Eye system categorizes network events into distinct alert types such as new device alerts, reconnection alerts, and disconnection alerts to ensure structured reporting and better interpretability. Each alert provides relevant details including the device MAC address, time of detection, and connection status, enabling clear identification of network changes. This categorization helps in prioritizing network events and quickly identifying unusual or unexpected device activity. For example, frequent reconnections from the same device or repeated appearance of unknown devices can be easily observed and analyzed. The structured alert system ensures that both real-time monitoring and post-analysis become more efficient and user-friendly.

## 4.4 Report Generation

The system is lightweight, efficient, and capable of real-time monitoring with minimal resource consumption, making it suitable for embedded platforms like the ESP32. It accurately tracks device connectivity and identifies key events such as new connections, reconnections, and disconnections without significant delay. The modular design allows easy expansion and integration with additional features such as alert systems, logging mechanisms, or advanced analytics tools. It performs reliably even in moderately crowded network environments by maintaining stable connections and structured monitoring. The system provides clear and organized output, making it easy to understand network activity during demonstrations and practical usage.

## 4.5 Discussion on System Behavior

### Strengths:

The system is lightweight, efficient, and capable of real-time monitoring with minimal resource consumption, making it suitable for embedded platforms like the ESP32. It accurately tracks device connectivity and identifies key events such as new connections, reconnections, and disconnections without significant delay. The modular design allows easy expansion and integration with additional features such as alert systems, logging mechanisms, or advanced analytics tools.

### Limitations:

The system monitors only devices connected to its own access point and does not analyze all nearby wireless networks. It relies solely on MAC address identification and does not perform deep packet inspection or traffic-level analysis. Performance may be affected in highly congested environments with multiple overlapping networks and interference. The number of devices that can be tracked is limited by the memory and processing capacity of the ESP32. .

### Future Enhancements:

Integration of external storage or cloud-based logging systems to enable long-term data collection and analysis. Implementation of alert mechanisms such as buzzers, notifications, or mobile integration for real-time warnings. Incorporation of advanced analytics techniques, including anomaly detection or machine learning, to improve detection accuracy. Enhancement of user interface through web dashboards or mobile applications for better visualization and control. Expansion of network capabilities to support more advanced routing features and improved scalability

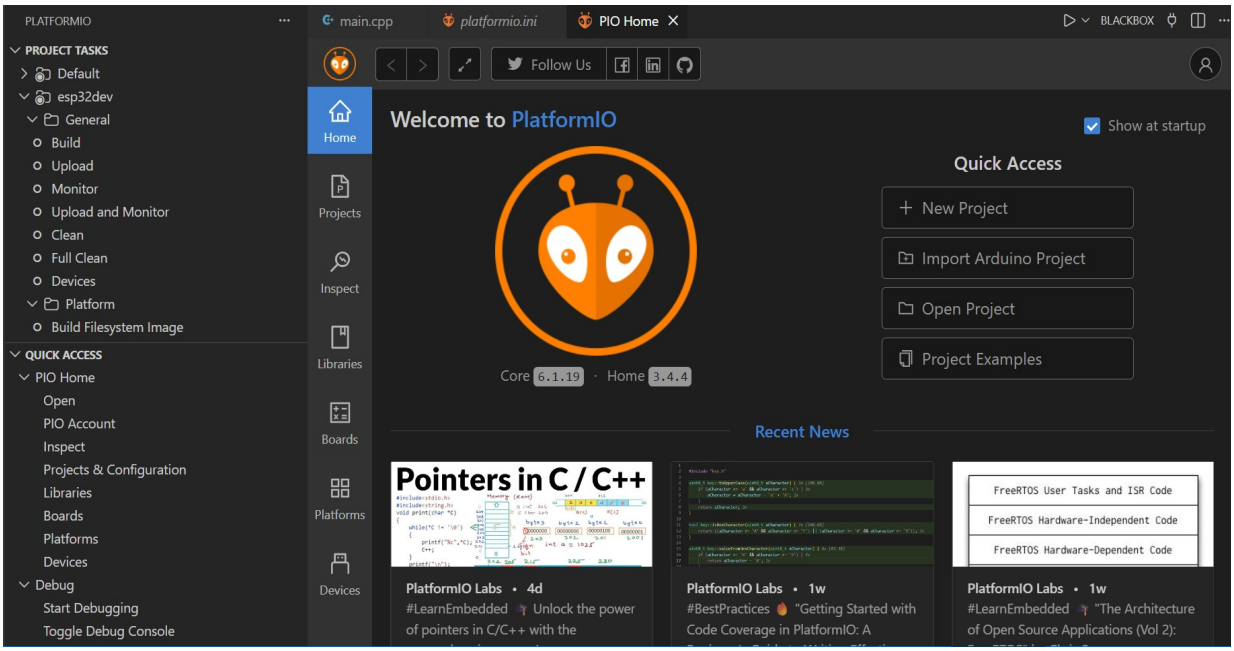
## 4.6 Overall Evaluation

The Hawk-Eye Wi-Fi Gateway system successfully achieved its objective of monitoring and analyzing device connectivity within a controlled wireless network environment. The system demonstrated reliable real-time detection of connected devices, accurately identifying new connections, reconnections, and disconnections without noticeable delay.

**Figure.1.1: The Panel for GUI of Mark 2 and Live Log**

**Viewer Table 2.1 Performance Metrics**

| Parameter | Performance / Observation |
|---|---|
| Device Detection Accuracy | ~95–100% (based on stable connections) |
| Real-Time Response | Near real-time (within 1–3 seconds delay) |
| Connection Stability | Moderate to High (depends on network) |
| Maximum Devices Tested | Up to 8–10 devices (stable performance) |
| Memory Usage Efficiency | Optimized for ESP32 constraints |
| Power Consumption | Low (suitable for continuous operation) |
| Detection Capability | New, Reconnect, Disconnect events |
| Scalability | Limited by hardware constraints |

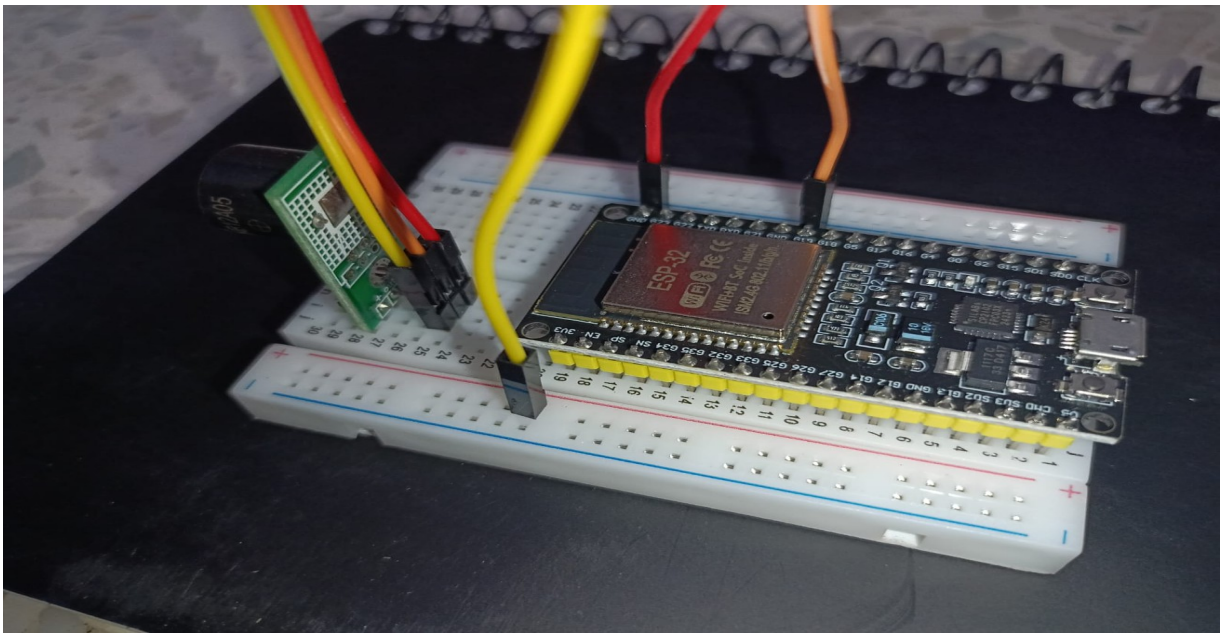


```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   Upload - Task

* Executing task: C:\Users\DELL\.platformio\penv\Scripts\platformio.exe run --target upload
Writing at 0x000607b6... (43 %)
Writing at 0x000659cc... (46 %)
Writing at 0x0006b5d4... (50 %)
Writing at 0x00070dde... (53 %)
Writing at 0x00075fe5... (56 %)
Writing at 0x0007b3dc... (60 %)
Writing at 0x0008062f... (63 %)
Writing at 0x00085da5... (66 %)
Writing at 0x0008b546... (70 %)
Writing at 0x0009108d... (73 %)
Writing at 0x00096afd... (76 %)
Writing at 0x0009c54d... (80 %)
Writing at 0x000a593e... (83 %)
Writing at 0x000acbbc... (86 %)
Writing at 0x000b20e6... (90 %)
Writing at 0x000b7dfe... (93 %)
Writing at 0x000bd16e... (96 %)
Writing at 0x000c2d6f... (100 %)
Wrote 733264 bytes (475648 compressed) at 0x00010000 in 11.7 seconds (effective 500.0 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
============================================ [SUCCESS] Took 32.24 seconds ============================================
```

THE SOURCE CODE HAS SUCCESSFULLY UPLOADED INTO ESP32

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   Upload...
                                                                                       Monitor...

* Executing task: C:\Users\DELL\.platformio\penv\Scripts\platformio.exe device monitor
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4
=== Hawk-Eye Phase 2 ===
Connecting to upstream hotspot...
.....
✅ Connected to Internet Source
IP: 10.34.206.67
Hawk-Eye Network Ready
Active Devices: 0
----------------------
Active Devices: 0
----------------------
[NEW]
MAC: 22:ED:FC:E2:D0:A5
Active Devices: 1
----------------------
Active Devices: 1
----------------------
Active Devices: 1
----------------------
Active Devices: 1
----------------------
```

WE CAN SEE THAT ESP32 CREATED ITS OWN HOTSPOT AND A DEVICE IS CONNECTED AND ITS MAC ID IS BEING DISPLAYED




THIS IS THE HARDWARE SETUP OF THIS PROJECT WHICH CONTAINS ESP32 , BREADBOARD , 5V ACTIVE BUZZER AND JUMPER CABLES

# CHAPTER 5
# CONCLUSION

# CHAPTER 5

# CONCLUSION

The Hawk-Eye project has successfully demonstrated a practical, efficient, and reliable approach to monitoring device connectivity within a wireless network using an embedded system. The primary objective of the project was to design a smart Wi-Fi gateway capable of tracking connected devices in real time, identifying connection patterns, and providing structured insights into network activity. This objective has been effectively achieved through the implementation of a dual-mode ESP32 system that operates both as an access point and as a station connected to an external network.

The core strength of this project lies in its ability to accurately detect and monitor devices using their MAC addresses while maintaining real-time responsiveness. The system efficiently identifies new device connections, recognizes previously known devices upon reconnection, and detects disconnections with minimal delay. This enables continuous tracking of network activity and provides a clear understanding of how devices interact within a controlled environment. The structured output generated through the serial interface ensures that all events are clearly presented, making the system easy to interpret and suitable for demonstration as well as analysis.

Another significant outcome of this project is the development of a modular and scalable design. The system is divided into distinct functional components such as access point management, device monitoring, state analysis, and output generation, ensuring clarity and ease of extension. This modular architecture allows future enhancements such as alert systems, data logging, cloud integration, or advanced analytics to be incorporated without major redesign. The lightweight nature of the implementation ensures that the system operates efficiently on the ESP32 without excessive resource consumption, making it suitable for continuous operation. During the development process, several challenges were encountered, particularly in handling unstable network conditions, managing device tracking in dynamic environments, and ensuring accurate differentiation between new and previously known devices.

Issues related to connection reliability and hotspot compatibility were addressed through retry mechanisms and proper configuration adjustments. These challenges contributed to refining the system and improving its overall robustness and stability. The performance evaluation of the system indicates

deep packet inspection, the system effectively fulfills its intended purpose as a lightweight network monitoring solution.Beyond its technical implementation, the project highlights the potential of embedded systems like ESP32 in the field of basic network monitoring and cybersecurity applications. It demonstrates how low-cost hardware combined with efficient programming can produce meaningful and practical solutions without requiring complex infrastructure. The project serves as a strong foundation for further exploration in areas such as smart networking, IoT security, and real-time monitoring systems. Looking forward, several enhancements can be considered to improve the system's capabilities. Integration of external storage or cloud-based logging would enable long-term data analysis and centralized monitoring. Implementation of real-time alert mechanisms such as buzzers, notifications, or mobile integration would increase responsiveness to critical events. Advanced techniques such as anomaly detection or machine learning could further enhance the system's ability to identify unusual patterns in network behavior. Additionally, the development of a graphical user interface or web-based dashboard could significantly improve usability and visualization. In conclusion, the Hawk-Eye system successfully achieves its goal of providing a smart and efficient Wi-Fi monitoring solution using an embedded platform. It bridges the gap between basic network connectivity and real-time device tracking, offering a scalable and modular approach that can be extended for more advanced applications. The project stands as a practical demonstration of how embedded systems and networking concepts can be combined to address real-world challenges, making it a valuable contribution in the domain of wireless monitoring and security.

The Hawk-Eye project has successfully demonstrated a practical, efficient, and reliable approach to monitoring device connectivity within a wireless network using an embedded system. The primary objective of the project was to design a smart Wi-Fi gateway capable of tracking connected devices in real time, identifying connection patterns, and providing structured insights into network activity. This objective has been effectively achieved through the implementation of a dual-mode ESP32 system that operates both as an access point and as a station connected to an external network, thereby simulating a gateway-like environment. The near real-time response of the system ensures minimal delay between an event occurring and its detection, which is essential for monitoring dynamic wireless environments. A key strength of the Hawk-Eye system lies in its simplicity combined with functionality.

# REFERENCES

# REFERENCES

[1] Espressif Systems, "ESP32 Technical Reference Manual," [Online]. Available: https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf

[2] Espressif Systems, "ESP-IDF Programming Guide," [Online]. Available: https://docs.espressif.com/projects/esp-idf/en/latest/esp32/

[3] Arduino IDE Documentation, "WiFi Library for ESP32," [Online]. Available: https://www.arduino.cc/reference/en/libraries/wifi/

[4] PlatformIO, "ESP32 Development with PlatformIO," [Online]. Available: https://docs.platformio.org/en/latest/platforms/espressif32.html

[5] IEEE, "Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications," IEEE 802.11 Standard.

[6] GeeksforGeeks, "Introduction to Wi-Fi and Networking Basics," [Online]. Available: https://www.geeksforgeeks.org/computer-network-tutorials/

[7] Random Nerd Tutorials, "ESP32 Access Point and Station Mode Guide," [Online]. Available: https://randomnerdtutorials.com/esp32-access-point-ap-web-server/

[8] Stack Overflow, "ESP32 Wi-Fi Programming and Troubleshooting Discussions," [Online]. Available: https://stackoverflow.com/questions/tagged/esp32

[9] GitHub, "ESP32 Wi-Fi Projects and Examples," [Online]. Available: https://github.com/search?q=esp32+wifi

[10] Computer Networking: A Top-Down Approach by James F. Kurose and Keith W. Ross, Pearson Education, 2017.

**GitHub Link**

1. https://github.com/JARVIS-1729